Treatment of anxiety in older adolescents and young adults with autism spectrum disorders: A pilot study

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Anxiety disorders are commonly comorbid in adolescents and young adults with high-functioning autism. Cognitive-behavioral treatments (CBT) for anxiety, when adapted and expanded to target autism spectrum disorder (ASD) characteristics, may be beneficial, but there is minimal evidence to guide clinicians in their application. This multiple-baseline design study evaluated the initial efficacy of a CBT protocol adapted to address anxiety symptoms and adaptive functioning in this population. Anxiety and ASD symptoms were assessed for six participants at intake, after baseline, posttreatment, and at 1-month follow-up. Parent- and child-reported anxiety was also assessed during baseline and treatment. Visual inspection and reliable change index scores were used to evaluate change. All participants improved on clinician-rated measures of disorder severity, and gains were maintained at follow-up. Results were more equivocal for parent- and self-rated anxiety and parent-rated ASD, partly because of spontaneous changes during baseline. (Bulletin of the Menninger Clinic, 84[2], 105–136)

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Keywords: anxiety, autism spectrum disorder, adolescents, young adults, cognitive-behavioral therapy

Autism spectrum disorders (ASD) affect as many as one out of every 68 children in the United States (Centers for Disease Control and Prevention [CDC], 2014) and comprise one of the most common neurobiological conditions in youth (CDC, 2012; Fombonne, 2005; Rutter, 2005). The increase in reported prevalence of ASD diagnoses over the past several years (CDC, 2009; Ouellette-Kuntz, et al., 2007) may reflect, in part, a broadening of the diagnostic criteria to include recognition of higher-functioning individuals (Fombonne, 2005; Rutter, 2005). Such higher-functioning individuals with an ASD, in contrast to lower-functioning individuals, generally have at least normative intelligence and syntactical speech (Szatmari et al., 2009), with difficulties likely to be apparent in pragmatic language, social communication and interactions, and restricted, repetitive, or stereotyped patterns of behaviors, interests, or activities. Higher-functioning individuals are often not diagnosed until school age or later, if ever, and some remain undiagnosed well into adulthood (Barnard, Harvey, Prior, & Potter, 2001; Woodbury-Smith, Robinson, Wheelwright, & Baron-Cohen, 2005).

Although higher intellectual functioning and adequate communication skills during early development are associated with better functioning and prognosis for those with ASD (Black, Wallace, Sokoloff, & Kentworthy, 2009; Fein et al., 1999), ex-

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isting research suggests that many youth with this condition will continue to experience difficulty throughout adolescence and adulthood (Barnard et al., 2001; Billstedt, Gillberg, & Gillberg, 2005). For example, difficulties with independence, obtaining gainful employment, and establishing relationships are commonly noted among adults with ASD diagnoses (Barnard et al., 2001; Morde et al., 2012). Thus, it would appear that late adolescence and young adulthood constitute an opportune time for interventions aimed at improving psychosocial or adaptive functioning in individuals with ASD.

Comorbid psychological disorders are common among adolescents and adults with ASD diagnoses, particularly internalizing disorders such as anxiety and depression (Ghaziuddin, Weidmer-Mikhail, & Ghaziuddin, 1998; Kim, Szatmari, Bryson, Streiner, & Wilson, 2000; Leyfer et al., 2006). Several studies have indicated that youth with ASD evidence higher levels of anxiety and depression relative to typically developing peers (Bellini, 2004; Hurtig et al., 2009; Kim et al., 2000; Kuusikko et al., 2008; Russell & Sofronoff, 2005). Anxious adolescents with and without an ASD diagnosis may experience similar severity of their anxiety symptoms (Farrugia & Hudson, 2006; Russell & Sofronoff, 2005). Farrugia and Hudson (2006) compared adolescents (ages 12–16 years) with Asperger's syndrome to two groups of adolescents: those with a clinical diagnosis of an anxiety disorder, and a nonclinical control group. Results revealed that adolescents with Asperger's had levels of anxiety equivalent to those experienced by the adolescents with diagnosed anxiety disorders, and their level of anxiety was significantly higher than that of the nonclinical control group (Farrugia & Hudson, 2006).

Even among typically developing youth, anxiety can negatively impact academic performance, peer relationships, and family cohesion (S. Barrett & Huebeck, 2000; Langley, Bergman, McCracken, & Piacentini, 2004). Anxiety may be particularly debilitating to individuals with an ASD diagnosis as a result of continuing social and emotional impairments associated with ASD, including deficits in social skills and competence, low levels of peer acceptance, and feelings of loneliness (Kuusikko et al., 2008; Lasgaard, Nielsen, Eriksen, & Goossens, 2010; Williamson, Craig, & Slinger, 2008). Furthermore, as they transition to adulthood, adolescents with ASD often lack the community connections, friendships and self-advocacy skills that facilitate this transition among typically developing youth (Baxter, 1997). Thus, adolescents and young adults with comorbid anxiety disorders and ASD are a vulnerable, understudied population for which potentially effective, cognitive-behavioral interventions require new research to support their comparative utility and implementation.

Evidence-based, psychosocial treatments have repeatedly demonstrated efficacy in the treatment of adolescents with a variety of internalizing disorders. Cognitive-behavioral therapy (CBT) is efficacious for various anxiety disorders, with effect sizes in the medium to large range reported in several meta-analyses (Higa-McMillan, Francis, Rith-Najarian, & Chorpita, 2016; Ishikawa, Okajima, Matsuoka, & Sakano, 2007). Although multiple CBT-oriented protocols exist for anxiety disorders in youth, these treatments typically employ various combinations of roughly similar procedures, most often utilizing exposure, modeling, operant, family, and/or cognitive-based procedures, and may be presented in either a group or individual format with roughly equivalent efficacy (P. M. Barrett, Dadds, & Rapee, 1996; Kendall, 1994; March, Mulle, & Herbel, 1994).

There is evidence to suggest that CBT may be beneficial in addressing the anxiety symptoms that are associated with, or comorbid with, ASD in youth (Chalfant, Rapee, & Carroll, 2007; Sofronoff, Attwood, & Hinton, 2005; Sze & Wood, 2007; Wood et al., 2015). However, to increase the likelihood of generalizability and long-lasting impact of CBT-related skills and knowledge in those with ASD, traditional CBT protocols have often been adapted and expanded to target ASD characteristics that may cause or compound anxiety symptoms. These characteristics include deficits in social skills and perspective taking, poor adaptive skills, and circumscribed interests and stereotypies (Wood, Drahota, & Sze, 2007). One such protocol that has been developed for children with ASD (Behavioral Intervention for Anxiety in Children with Autism [BIACA]; Wood et al., 2007) has shown promising evidence of significant improvement in anxiety symptoms and ASD-specific areas of impairment such as social initiation and self-help skills. Furthermore, this modified form of CBT has demonstrated efficacy in dealing with stereotyped behaviors, perseveration, social isolation, and reciprocal communication (Sze & Wood, 2007).

Although efficacious with younger children with ASD, CBT for older adolescents and young adults with ASD may require developmentally appropriate modifications to address issues of emerging independence, unique social skill and self-advocacy needs, and life transitions to be fully appropriate for this population. One such protocol that has been informed by developmental considerations relevant to adolescents is the Multimodal Anxiety and Social Skills Intervention (MASSI; White et al., 2010), a CBT-based program for adolescents with ASD that incorporates social skills training and includes individual, group, and parent components. An initial pilot randomized controlled trial (RCT) examining the feasibility and preliminary efficacy of MASSI in 30 early adolescents demonstrated a statistically significant improvement for MASSI participants, with a within-group effect size of $d = 1.18$. In addition, greater improvements with respect to both overall anxiety and global functioning were noted in the MASSI group as compared to a waitlist control group (White et al., 2013). Although results provide evidence of the feasibility and efficacy of MASSI, this protocol is geared specifically toward adolescents and may not address the unique social and role challenges faced by individuals with ASD who are transitioning to young adulthood. To fill this gap in the literature, the efficacy of a 16-week modularized CBT intervention for late adolescents with anxiety and high-functioning autism was tested in a sample of six late adolescents/early adults with ASD and comorbid anxiety (Treatment of Anxiety in Late Adolescents With Autism [TALAA]; Wise et al., 2019). Results revealed significant improvements in clinician-rated global severity and anxiety symptoms from baseline to posttreatment, but not participant-rated anxiety (Wise et al., 2019). While results of this early open trial were promising, further investigation of this type of protocol is warranted due to the lack of a controlled baseline phase in the study. In addition, investigators did not examine change in parent report of symptoms, which are important to consider in this population into early adulthood due to the increased role of parents in the lives of these individuals.

The current study examined the initial effects of a modified CBT program for youth (i.e., Wood et al., 2007) to address anxiety and other key concerns for adolescents and young adults with ASD. This modified CBT treatment protocol utilized elements of an evidence-based social skills training protocol for adolescents with ASD (i.e., Program for the Education & Enrichment of Relational Skills [PEERS]; Laugeson & Frankel, 2010), modules focusing on motivation and behavioral activation to address depression symptoms more common in this age group (Merikangas & Knight, 2009), and new materials focused on appropriate, ideographically defined role transitions (e.g., school to work). Our aim was to evaluate the initial efficacy of this protocol on symptoms of anxiety, depression, and autism using a multiple-baseline design.

Method

Research design

This pilot study utilized a single-case, multiple baseline across subjects design (Barlow, Nock, & Hersen, 2009; Kazdin, 2003) to evaluate the feasibility of the adapted intervention protocol for addressing anxiety symptoms and adaptive functioning in youth with ASD. Institutional review board approval was obtained from the institution at which the study was conducted before beginning any study procedures. Participants were randomly assigned to a 4-week, 8-week, or 12-week waitlist condition to allow for the collection of baseline data prior to the start of the treatment protocol. This baseline period consisted of biweekly phone check-in sessions, which provided a control condition to which the impact of the weekly intervention could be compared. This design also allowed for an assessment of change in the rate of symptom improvement during treatment relative to baseline within and between subjects (Barlow & Nock, 2009).

Participants

This pilot study sample included six adolescents and young adults with ASD and an anxiety disorder, ranging in age from 15 to 21 years (see Table 1 for demographic information). There

Table 1. **Participant demographics**

P	Age	Gender	Baseline Length (weeks)	# Sessions Attended	Baseline SCQ	Anxiety Disorder (CSR)			
						Baseline	Postbaseline	Posttreatment	1-Month Follow-Up
1	15	M	8	16	7	GAD (6)	GAD (6)	GAD (4)	GAD (4)
						SOC (4)	SOC (5)	SOC (3)	SOC (2)
2	14	M	12	14	19	SOC (5)	SOC (5)	SOC (4)	SOC (4)
						ANX-NOS (4)	ANX-NOS (4)	ANX-NOS (2)	ANX-NOS (2)
						SP (2)	SP (2)	SP (1)	
3	15	M	12	16	N/A	GAD (6)	GAD (6)	GAD (3)	GAD (2)
						SOC (4)	SOC (5)	SOC (4)	SOC (4)
4	20	M	4	16	19	GAD (6)	GAD (5)	GAD (3)	GAD (4)
						SOC (4)	SOC (6)	SOC (3)	SOC (4)
						SP (3)	SP (2)	SP (1)	SP (2)
5	18	M	8	16	18	GAD (5)	GAD (4)	GAD (2)	N/A
						SOC (4)	SOC (5)	SOC (3)	N/A
6	17	F	4	11	20	GAD (6)	GAD (6)	GAD (5)	N/A
						SOC (5)	SOC (5)	SOC (4)	N/A
						MDD (4)	MDD (4)	MDD (4)	N/A

Note. P: participant; SCQ: Social Communication Questionnaire; CSR: Clinical Severity Rating; GAD: generalized anxiety disorder; SOC: social phobia; SP: specific phobia; ANX-NOS: anxiety disorder not otherwise specified; MDD: major depressive disorder.

were five males in this sample: Two were in the 9th grade, attending mainstream high school classes (both 15 years of age); one was in 11th grade at a private, religious school (16 years of age); one was in his senior year of high school (18 years of age); and one was completing his associate's degree at a two-year community college (21 years of age). There was one female in the sample who was completing 11th grade and attending mainstream high school classes (17 years of age). All of the sample identified as White, and 50% of those (n = 3) also identified as Hispanic/Latino. All participants in the study were living in the same home with the parent/caregiver who participated in treatment.

Inclusion criteria for this study were a clinical diagnosis of an ASD within the past 6 months, or meeting the autism or ASD cutoff score on the Autism Diagnostic Observation Schedule (Lord, Rutter, DiLavore, & Risi, 2002); being between the ages of 14 and 22 years old at time of intake assessment; and meeting diagnostic criteria for an anxiety disorder, as determined based on an administration of the Anxiety Disorders Interview Schedule for *DSM-IV*, Child Version (Silverman & Albano, 1996). Adolescents and young adults were included if they were functioning at or above grade level and/or had a documented IQ greater than 70, to allow for an understanding of the assent process as well as full participation in the cognitive components of the protocol. Moreover, if the adolescent/adult was on any psychiatric medication, a 1-month stabilization period (for benzodiazepines) or 3-month stabilization period (for selective serotonin reuptake inhibitors [SSRIs], serotonin and norepinephrine reuptake inhibitors [SNRIs], or tricyclics) was required prior to the initial diagnostic assessment. All participants were asked to have one parent (or caregiver with whom the adolescent or young adult was living) available to accompany them to assessment sessions and participate in caregiver-directed sessions as prescribed. Parents gave written informed consent and adolescents gave written assent (unless the individual was 18 years of age or older, in which case the participant provided written consent and a release of information regarding caregiver involvement).

Participants were excluded if there was a current or past diagnosis of schizophrenia, bipolar I or II disorder, organic brain syndrome, intellectual disability, or current suicidal/homicidal ideation. Participants with other types of comorbid conditions (e.g., substance use disorders or disruptive behavior disorders) were not excluded, provided their principal diagnosis was an anxiety disorder. Adolescents and their parents had to be able to speak, read, and understand English sufficiently well to complete study procedures.

Measures

Anxiety Disorders Interview Schedule for DSM-IV, Child Version, Child and Parent Report Forms (ADIS-IV-C/P). The ADIS-IV-C/P (Silverman & Albano, 1996) is a semistructured interview permitting the diagnosis of all *DSM-IV* (American Psychiatric Association, 2000) anxiety disorders, mood disorders, and externalizing disorders of childhood and adolescence, and providing screening questions for selected other disorders. Although usage of the adult version of the ADIS was considered for participants over age 18, we determined that instrument consistency and parental involvement were of primary importance for this pilot trial. On the basis of youth and caregiver reports, evaluators assign a Clinical Severity Rating (CSR) score for each diagnosis ranging from 0 to 8, with "0" reflecting no symptoms/functional impairment and "8" reflecting marked severity and impairment. Psychometric properties for the *DSM-IV* version of the ADIS-C/P have shown adequate test–retest reliability for anxiety and depressive disorders, with a test–retest interval of 7–14 days (Silverman, Saavedra & Pina, 2001). The full ADIS-IV-C/P was used during the initial diagnostic assessment in the present study, but a shorter version of the ADIS-IV-C/P (i.e., the Mini-ADIS-IV-C/P) was used after the baseline period, at posttreatment, and at the 1-month follow-up assessment. This version focuses on the interim period since the previous assessment and was administered by an independent evaluator blind to treatment status. The ADIS-IV-C/P was administered by one PhD-level clinician and one predoctoral psychology graduate student who received supervision on each case from an expert rater. Neither clinician provided treatment to study participants.

Autism Diagnostic Observation Schedule (ADOS). The ADOS (Lord et al., 2002) is a semistructured observational assessment for the diagnosis of autism that measures communication, social interaction, play, and the imaginative use of materials. The structured activities have been divided into four modules appropriate to children and adults at different developmental linguistic levels. A recent meta-analysis found that, across five studies, the ADOS demonstrated a correct classification score for autism of .80, with some variability depending upon the module and scoring algorithm used (Falkmer, Anderson, Falkmer, & Horlin, 2013). Modules 3 and 4 were used in this study, and the assessment was only given at intake by a research-reliable examiner if there was no prior ASD diagnosis.

Revised Child Anxiety and Depression Scale–Parent Version (RCADS-P). The RCADS-P (Ebesutani, Bernstein, Nakamura, Chorpita, & Weisz, 2010) is a 47-item parent-report questionnaire with scales corresponding to separation anxiety disorder (SAD), social phobia (SOC), generalized anxiety disorder (GAD), panic disorder (PD), obsessive-compulsive disorder (OCD), and major depressive disorder (MDD). Respondents rate how often each item applies to their child on a scale from 0 to 3, corresponding to "never," "sometimes," "often," and "always." The RCADS-P has shown acceptable to good internal consistency in clinic-referred youth (with Cronbach's alpha for individual subscales ranging from .68 to .84), as well as favorable convergent, discriminant, and factorial validity (Ebesutani et al., 2010). The RCADS-P was completed at each assessment point, as well as during each baseline phone check-in and each treatment session. For the purposes of this investigation, scores on the GAD, SOC, and Total Anxiety scales were analyzed.

Behavior Assessment System for Children, Second Edition, Parent Rating Scale–Adolescent Version, and Self-Report of Personality–Adolescent Version (BASC-2-PRS-A and BASC-2-SRP-A). The BASC-2 system (Reynolds & Kamphaus, 2004) assesses emotional, behavioral, and adaptive functioning of adolescents aged 12 to 21 years. Subscales address a broad range of both internalizing and externalizing concerns. The BASC-2-PRS-A and BASC-2-SRP-A have been standardized and normed on large national samples, and all of the scales have well established reliability and validity (Reynolds & Kamphaus, 2004). The BASC-2-PRS-A and BASC-2-SRP-A were completed at each assessment time point (i.e., intake, postbaseline, post-treatment, and 1-month follow-up), and scores on the Anxiety and the Internalizing Problems Composite scales from both parent and adolescent versions were examined.

Social Responsiveness Scale (SRS). The SRS (Constantino & Gruber 2005) is a 65-item parent checklist of items relating to children's social ability and impairment in naturalistic social settings. The SRS has been designed to provide a quantitative index of social behavior that is sensitive across the full range of typical and atypical social development, as well as sensitive to distinctions between pervasive developmental disability samples and other clinical groups. Each item is rated on a scale from "never true" (0) to "almost always true" (3), with higher scores indicating more social impairment. The measure has excellent short- and longer-term test–retest reliability (.83 to .88) and convergent multiple informant agreement (Constantino, Przybeck, Friesen, & Todd, 2000; Constantino & Todd, 2003; Constantino et al., 2003). The SRS was completed at each assessment time point, and the SRS Total score was examined.

Social Communication Questionnaire (SCQ). The SCQ (Berument, Rutter, Lord, Pickles, & Bailey, 1999) is a brief, parent-report instrument for the valid screening or verification of autism spectrum disorder symptoms in youth. The total score represents the three critical autism diagnostic domains of qualitative impairments in reciprocal social interaction, communication, and repetitive and stereotyped pattern behaviors; 15 is the suggested cutoff score for an autism diagnosis. The SCQ demonstrates good reliability and validity, with coefficient alphas for various age groupings ranging from .84 to .93 (Berument et al., 1999). This measure was administered at intake as a measure of autism severity. Participant scores ranged from 7 to 20, with four of the five participants who completed the measure falling above the cutoff of 15. See Table 1 for further information about participant scores.

Intervention

The Anxiety-Focused Intervention for Adolescents and Young Adults with Autism (AFIYA) is a modified version of the Behavioral Intervention for Anxiety in Children with Autism (BIACA; Wood et al., 2007) similar to the intervention protocol used in a recently published open trial of modular CBT for late adolescents with anxiety and ASD (Wise et al., 2019). Relative to BIACA, the AFIYA protocol places greater emphasis on the cognitive components of anxiety, uses motivational enhancement and behavioral activation (to address conduct issues and depressive symptoms), uses photographs for emotion education, and uses role-play and video-modeling techniques to teach both self and other emotional awareness and social skill development. In addition, there is an emphasis on developing independence, autonomy, and both basic and more sophisticated self-advocacy and social and communication skills in the context of major life transitions. Source materials from the PEERS (Laugeson & Frankel, 2010), as well as other cognitive-behavioral manuals for anxiety and depression in youth and adults, were also incorporated (Barlow et al., 2017; Ehrenreich-May et al., 2018).

AFIYA is a modular cognitive-behavioral intervention that consists of approximately 16 weekly treatment sessions. These sessions last 90 minutes and include sections dedicated to speaking with the adolescent or young adult alone, the caregiver alone, and the caregiver and adolescent/young adult together. The program's flexible modular structure allows clinicians to individually tailor treatment by selecting specific modules that best meet the needs of clients and their families. Modules are selected on the basis of a simple algorithm adapted from Chorpita, Taylor, Francis, Moffitt, and Austin (2004). The algorithm allows clinicians to deliver basic coping skills by using the KICK steps (Knowing I'm Nervous; Irritating Thoughts; Calm/Challenge Thoughts; Keep Practicing), developmentally tailored using adolescent/young adult friendly examples and language, followed by a primary focus on exposures. Supplementary modules may be used to address barriers that occur during exposures or to provide social skills training. Parent modules also follow an algorithmic structure and include modules focused on psychoeducation, encouraging independence and related self-advocacy skills, helping their adolescents/young adults to negotiate

Table 2. **AFIYA treatment modules and participants receiving each module**

Adolescent Module	Module Contents	P1	P2	P3	P4	P5	P6
1	Introduction, Information about ASD	X	X	X	X	X	X
2	"K" and "I" Steps	X	X	X	X	X	X
3	"C" Steps	X	X	X	X	X	X
H	Hierarchy of Exposures	X	X	X	X	X	X
4	"K"- Keep Practicing	X	X	X	X	X	X
KICK	Using the KICK Plan	X	X	X	X	X	X
IV	In Vivo Exposure	X	X	X	X	X	X
R	Relaxation	X	X	X	X	X	
ENTRY/EXIT	Entering/Exiting Social Interactions	X	X	X	X	X	
PLAY	Get-Togethers/Friendships	X	X	X	X	X	
ERP	E/RP In Vivo Exposure						
T	Termination	X	X	X	X	X	X
Parent Module							
I	Introduction, Information, and Treatment Plan	X	X	X	X	X	X
IND	Encouraging Independence					X	
H	Hierarchy of Exposures	X	X	X	X	X	X
EXP	Negotiating Exposures						
REW	Rewards						
APPROP	Socially Appropriate Interaction and Activities						
T	Termination	X	X	X	X	X	X

Note. KICK: Knowing I'm Nervous; Irritating Thoughts; Calm/Challenge Thoughts; Keep Practicing.

exposures, and facilitating age-appropriate social interactions. A complete list of all modules, as well as information about which participants received which modules, may be found in Table 2.

Analytic strategy

Changes in symptomatology and functioning over time were examined by visually inspecting patterns of change on the parent-report and adolescent-report measures, as well as clinicians' ratings of severity and functioning as indicated by the ADIS-IV-C/P

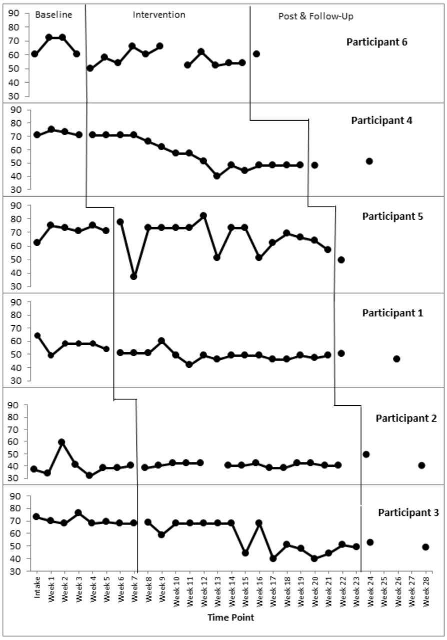

Figure 1. Weekly RCADS-P Social Anxiety ***t*** scores across baseline, intervention, and posttreatment and follow-up phases for all participants. Baseline phase time points include both intake assessment and assessment at the end of the baseline period.

CSR scores across assessment points (Hayes, Barlow, & Nelson-Gray, 1999; Kazdin, 2003). Changes that are large in magnitude, temporally related to the baseline-to-intervention phase change, consistent throughout the intervention phase, and similar across participants allow for the strongest conclusions to be drawn about the relationship between the intervention and reported symptomatology. Table 1 summarizes changes in diagnostic status and related severity over time. Figure 1 provides a graphical representation of changes in parent-rated social anxiety symptoms during the baseline and treatment phases for each participant, while Figure 2 provides a graphical representation of changes in parent-rated generalized anxiety symptoms during the baseline and treatment phases for reach participant.

Reliable change index scores (RCI) were also calculated to determine the significance of changes in the RCADS-P, BASC-2-PRS-A, BASC-2-SRP-A, and SRS scores during the waitlist and intervention phases above and beyond that of normal measurement

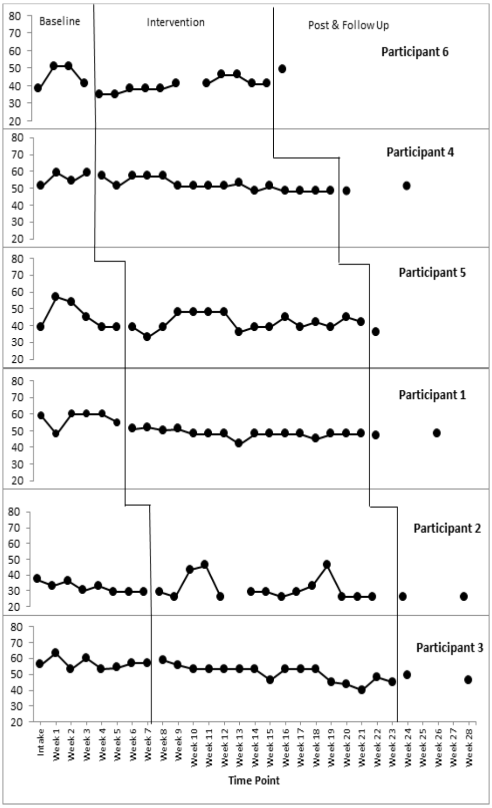

Figure 2. Weekly RCADS-P Generalized Anxiety ***t*** scores across baseline, intervention, and posttreatment and follow-up phases for all participants. Baseline phase time points include both intake assessment and assessment at the end of the baseline period.

variance and/or change due to the passage of time. Specifically, RCI scores were calculated for each adolescent on each scale for three time points: (1) intake to the end of each participant's baseline period (hereafter referred to as postbaseline), (2) postbaseline to completion of the intervention (hereafter referred to as posttreatment), and (3) posttreatment to a 1-month follow-up. RCIs larger than the *z*-score level of significance, in this case 1.96 ($p < .05$), were considered statistically reliable (Ferguson, Robinson, & Splaine, 2002). Table 3 summarizes posttreatment reliable change scores for all participants on all measures, as well as whether gains were maintained at follow-up.

Results

Participant 1 (P1)

Changes in diagnostic status. At intake, P1 (15-year-old male, 8-week baseline) met diagnostic criteria for generalized anxiety disorder (GAD; CSR = 6) and social phobia (SOC; CSR = 4). From intake to postbaseline, P1's CSR score remained stable for GAD (CSR = 6), but symptom severity for SOC was judged to have worsened slightly (CSR = 5). From postbaseline to posttreatment, decreases in CSR scores were reported for both domains (GAD CSR = 4; SOC CSR = 3).Treatment gains were maintained at 1-month follow-up, with P1 continuing to have a CSR of 4 for GAD and a CSR of 2 for SOC.

Changes in parent report of emotional symptoms. P1 demonstrated a statistically reliable decrease on the RCADS-P Total Anxiety scale from intake to postbaseline (RCI = –2.14), although the postbaseline score remained in the clinically significant range ($t \geq 65$). Visual inspection revealed a decrease on the RCADS-P-SOC scale that was not statistically reliable (RCI = –1.43), and there was little change on the RCADS-P GAD scale (RCI = –.45). From postbaseline to posttreatment, visual inspection indicated decreases on the RCADS-P Total Anxiety scale, the RCADS-P GAD scale, and the RCADS-P SOC scale, although these changes were not statistically reliable. All three RCADS-P scales remained stable at follow-up. There were no statistically reliable changes on the BASC-2-PRS-A during the baseline period on either the Anxiety (RCI = –.33) or the Internalizing Problems Composite scale (RCI = –.91). From postbaseline to posttreatment, statistically reliable changes were noted on both the Anxiety and Internalizing Problems Composite scales. It should also be noted that P1's scores on both scales decreased from the clinically significant range (i.e., $t > 65$) to the nonclinical range following treatment, and these decreases were maintained at the follow-up.

Changes in parent report of ASD-related symptoms. P1 demonstrated a statistically reliable decrease in symptoms on the SRS Total scale during the baseline period (RCI = –3.70), and

Table 3. **Posttreatment reliable change in outcome measures and maintenance of gains at 1-month follow-up**

	P1	FU	P2	FU	P3	FU	P4	FU	P5	FU	P6	FU
Pa RCADS Total	–1.79	—	–.18	—	–1.43	—	–3.22	Y	–2.50	Y	.54	—
Pa RCADS GAD	–1.34	—	0.00	—	–.89	—	–1.79	—	–.45	—	1.34	—
Pa RCADS Soc	–1.43	—	0.00	—	–2.51	Y	–4.30	Y	–3.59	Y	0.00	—
Pa BASC Anx	–3.62	Y	0.00	—	–3.62	Y	–1.15	—	–4.60	Y	.66	—
Pa BASC Int	–3.65	Y	0.00	—	–.36	—	–1.39	—	–3.88	Y	1.14	—
SRS	–.34	—	2.69	—	–.29	—	–9.43	Y	–6.73	Y	0.00	—
A BASC Anx	–.57	—	1.15	—	–2.68	Y	1.52	—	–1.15	—	.96	—
A BASC Int	–1.93	—	2.25	—	–3.21	N	0.00	—	–.96	—	–.96	—

Note. P: participant; FU: follow-Up; Pa: parent; A: adolescent; RCADS: Revised Child Anxiety and Depression Scale; BASC: Behavioral Assessment Scale for Children; SRS: Social Responsiveness Scale; GAD: generalized anxiety disorder; Soc: social anxiety disorder; Int: internalizing; Anx: anxiety. Scores that meet or exceed threshold of 1.96 for statistically reliable change are bolded. FU column indicates whether statistically reliable changes were maintained at FU (Y/N).

this improvement was maintained at the posttreatment and follow-up time points.

Changes in adolescent report of emotional symptoms. No reliable changes were noted during the baseline period on the BASC-2-SRP-A Anxiety scale (RCI = –.38) or the Internalizing Problems Composite scale (RCI = .64). From postbaseline to posttreatment, no reliable change was noted on the BASC-2-SRP-A Anxiety scale. Visual inspection revealed a decrease on the Internalizing Problems Composite scale, although this change was not statistically reliable. P1's scores on the BASC-2-SRP-A Anxiety and Internalizing Problems Composite scales remained stable at the 1-month follow-up.

Participant 2 (P2)

Changes in diagnostic status and clinicians' ratings. At intake, P2 (14-year-old male, 12-week baseline) met diagnostic criteria for SOC (CSR = 5) and anxiety disorder not otherwise specified (ANX-NOS; CSR = 4). Subclinical features of a specific phobia were also noted (SP; CSR = 2). No change in diagnostic status or CSR scores was observed from intake to postbaseline. All CSR scores declined posttreatment (SOC, CSR = 4; ANX-NOS, CSR = 2; SP, CSR = 1). These improvements were maintained at the 1-month follow-up (SOC, CSR = 4; ANX-NOS, CSR = 2; SP, CSR = 0).

Changes in parent report of emotional symptoms. During the baseline period, no reliable changes were observed in P2's score on the RCADS-P Total Anxiety scale (RCI = 1.07) or on the RCADS-P SOC scale (RCI = 1.79), although visual inspection revealed increases on both scales. P2's score on the RCADS-P GAD scale at intake was a zero. P2's score on the RCADS-P Total Anxiety, SOC, and GAD scales remained stable through the posttreatment and follow-up assessments. This pattern of findings mirrored that seen on the BASC-2-PRS-A Anxiety and Internalizing Problems Composite scales.

Changes in parent report of ASD-related symptoms. P2 displayed no reliable change in the SRS Total score from intake to postbaseline. At posttreatment, P2 showed a statistically reliable worsening of symptoms on the SRS Total scale. At follow-up, P2's SRS Total score decreased, although not to a statistically reliable degree.

Changes in adolescent report of emotional symptoms. From intake to postbaseline, P2 evidenced a decrease on the BASC-2-SRP-A Anxiety scale that was not statistically reliable (RCI = −1.53); however, there was a statistically reliable decrease on the Internalizing Problems Composite scale (RCI = −2.99). Posttreatment, there was no reliable change on the Anxiety scale, but there was a statistically reliable increase on the Internalizing Problems Composite scale. However, it should be noted that P2's posttreatment Internalizing Problems Composite scale score (*t* score = 42) was actually lower than his original intake score (*t* score = 44) due to a decrease during the baseline period. At follow-up, P2 evidenced no further statistically reliable changes on the Internalizing Problems Composite scale (RCI = −1.29) or the Anxiety scale (RCI = −.58).

Participant 3 (P3)

Changes in diagnostic status and clinicians' ratings. P3 (15-year-old male, 12-week baseline) met criteria for GAD (CSR = 6) and SOC (CSR = 4) at intake. His CSR for SOC increased to a 5 after the baseline period, suggesting a mild worsening of social anxiety symptoms. Posttreatment, decreases in CSR scores were observed for both diagnoses (GAD, CSR = 3; SOC, CSR = 4). At follow-up, the CSR score remained stable for SOC (CSR = 4), whereas the CSR for GAD had decreased slightly (CSR = 2).

Changes in parent report of emotional symptoms. During the baseline period, P3 evidenced a decrease on the RCADS-P Total Anxiety scale that did not reach statistical reliability (RCI = −1.79) and no change on the RCADS-P GAD scale (RCI = .00). P3's scores on the RCADS-P SOC scale decreased reliably during the baseline period (RCI = −2.15). From postbaseline to posttreatment, a further statistically reliable decrease was observed on the RCADS-P SOC scale, and a decline was observed on the RCADS-P Total Anxiety scale that did not reach statistical reliability. Treatment gains on the RCADS-P-SOC scale were maintained at follow-up. No statistically reliable changes were observed in P3's RCADS-P GAD or RCADS-P Total Anxiety scale at posttreatment or follow-up. During the baseline period, there were no reliable changes on the BASC-2-PRS-A Anxiety scale (RCI = 1.64) or the Internalizing Problems Composite scale (RCI = 1.60), although visual inspection revealed an increase on both composites. From postbaseline to posttreatment, a statistically reliable improvement was noted on the BASC-2-PRS-A Anxiety scale but not on the Internalizing Problems Composite scale. These improvements were maintained at the follow-up. In addition, as was the case for P1, P3's scores on the BASC-2-PRS Anxiety and Internalizing Problems Composite scales decreased from the clinically significant range to the nonclinical range posttreatment.

Changes in parent report of ASD-related symptoms. The SRS Total scale indicated no statistically reliable change from the intake to postbaseline assessment, or from the postbaseline to posttreatment assessment. However, P3 evidenced a statistically reliable decrease on the SRS Total scale (RCI = –4.04) at the follow-up period.

Changes in adolescent report of emotional symptoms. During the baseline period, P3 evidenced an increase in the BASC-2-SRP-A Anxiety scale that was not statistically reliable (RCI = 1.70) and a statistically reliable increase on the Internalizing Problems Composite scale (RCI = 3.21). Posttreatment, there was a reliable decrease on both the Anxiety and the Internalizing Problems Composite scales. Although P3's improvements on the Anxiety scale were maintained at follow-up, there was a statistically reliable increase in P3's overall Internalizing Problems Composite scale score (RCI = 5.15).

Participant 4 (P4)

Changes in diagnostic status and clinicians' ratings. P4 (20-year-old male, 4-week baseline) met criteria for GAD (CSR = 6) and SOC (CSR = 4) at intake and was given a subclinical diagnosis of specific phobia (SP; CSR = 3). Postbaseline, the CSRs for SP (CSR = 2) and GAD (CSR = 5) had decreased slightly; however, the CSR for SOC had increased (CSR = 6). Posttreatment, P4 showed reductions in CSR scores for GAD (CSR = 3), SOC (CSR = 3), and SP (CSR = 1). At 1-month follow up, slight increases in the CSR scores for SOC (CSR = 4), GAD (CSR = 4), and SP (CSR = 2) were noted.

Changes in parent report of emotional symptoms. There were no statistically reliable changes on P4's RCADS-P Total Anxiety scale (RCI = .71), RCADS-P SOC scale (RCI = .00), or RCADS-P GAD scale (RCI = 1.34) from intake to postbaseline. A statistically reliable decrease was noted on the RCADS-P Total Anxiety scale from postbaseline to posttreatment, and this improvement remained stable at follow-up. P4's score on the RCADS-P SOC scale also decreased reliably from postbaseline to posttreatment, and these gains were maintained at follow up (RCI = 1.08). Posttreatment, visual inspection also showed a decrease on the RCADS-P GAD scale, although this change was not statistically reliable. For the BASC-2-PRS, P4's results indicated that there were no statistically reliable changes on either the Anxiety or the Internalizing Problems Composite scales during the baseline period, at posttreatment, or at follow-up.

Changes in parent report of ASD-related symptoms. P4 showed a statistically reliable worsening of symptoms from intake to postbaseline on the SRS Total Scale (RCI = 5.39). Posttreatment, statistically reliable improvements were noted on the SRS Total scale; these gains were maintained at the 1-month follow-up.

Changes in adolescent report of emotional symptoms. During the baseline period, P4 evidenced a statistically reliable decrease on the BASC-2-SRP-A Anxiety scale (RCI = –2.68) but no reliable change on the Internalizing Problems Composite scale (RCI = –.032). At posttreatment, no reliable change was noted on the Anxiety scale or the Internalizing Problems Composite scale. However, at follow up, P4 demonstrated a reliable improvement on the Internalizing Problems Composite scale (RCI = –2.25).

Participant 5 (P5)

Changes in diagnostic status and clinicians' ratings. At intake, P5 (18-year-old male, 8-week baseline) met criteria for GAD (CSR = 5) and SOC (CSR = 4). Postbaseline, the CSR scores were reversed, with SOC judged to be more impairing (CSR = 5) than GAD (CSR = 4). Posttreatment, the CSR scores for SOC (CSR = 3) and GAD (CSR = 2) both decreased to subclinical levels. Follow-up data were not available for P5.

Changes in parent report of emotional symptoms. From intake to postbaseline, there were no statistically reliable changes in P5's scores on the RCADS-P Total Anxiety scale (RCI = 1.07), the RCADS-P GAD subscale (RCI = .00), or the RCADS-P SOC scale (RCI = 1.43). From postbaseline to post-treatment, P5 evidenced a statistically reliable decrease on the RCADS-P Total Anxiety scale and the RCADS-P SOC scale but not on the RCADS-P GAD subscale. On the BASC-2-PRS-A, P5's scores remained stable during the baseline period, and a reliable improvement was noted on both the Anxiety scale and the Internalizing Problems Composite scale from postbaseline to posttreatment. As was the case for P1 and P3, P4's level of symptoms on the BASC-2-PRS Anxiety and Internalizing Problems Composite scales decreased from the clinically significant (i.e., $t > 65$) range to the normal range during the intervention and were maintained during follow-up.

Changes in parent report of ASD-related symptoms. During the baseline period, there was no reliable changes on P5's scores on the SRS Total scale. From postbaseline to posttreatment, statistically reliable improvement was noted on the SRS Total scale, and improvement was maintained during follow-up.

Changes in adolescent report of emotional symptoms. From intake to postbaseline, no reliable changes were observed on the BASC-2-SRP-A Anxiety scale (RCI = 1.15) or the Internalizing Problems Composite scale (RCI = –1.29). Posttreatment, decreases on the Anxiety and the Internalizing Problems Composite scales were not statistically reliable.

Participant 6 (P6)

Changes in diagnostic status and clinicians' ratings. P6 (17-year-old female, 4-week baseline) was assigned diagnoses at intake of GAD (CSR = 6), SOC (CSR = 5), and major depressive disorder (MDD, CSR = 4). No changes in diagnostic status or CSR ratings were observed postbaseline. Posttreatment, slight improvements were noted in severity of GAD (CSR = 5) and SOC (CSR =4), whereas the CSR for MDD (CSR = 4) remained the same. Follow-up data were not available for P6.

Changes in parent report of emotional symptoms. During the baseline period, there were no statistically reliable changes in P6's scores on the RCADS-P Total Anxiety scale (RCI = –.71), the RCADS-P GAD subscale (RCI = .45), or the RCADS-P SOC scale (RCI = .00). No statistically reliable changes were noted posttreatment on any of the RCADS-P scales, the BASC-2-PRS-A Anxiety scale, or the Internalizing Problems Composite scale.

Changes in parent report of ASD-related symptoms. P6 did not evidence any statistically reliable changes in SRS-P Total score during the baseline period or following the intervention.

Changes in adolescent report of emotional symptoms. From intake to postbaseline, P6 evidenced a decrease on the BASC-2-PRS-A Anxiety scale that was not statistically reliable (RCI = –1.72) and a statistically reliable improvement on the Internalizing Problems Composite scale (RCI = –5.46). Posttreatment, there were no reliable changes on the Anxiety or the Internalizing Problems Composite scales.

Discussion

Research suggests that individuals with ASD may be more likely to experience clinically significant levels of anxiety compared to their typically developing peers (Bellini, 2004; Hurtig et al., 2009; Kim et al., 2000; Kuusikko et al., 2008; Russell & Sofronoff, 2005). It is widely accepted that without appropriate intervention, anxiety symptoms tend to persist over the life span (Pine & Grun, 1999). Difficulties with anxiety may further compound the social challenges experienced by individuals with ASD. Adolescence and early adulthood constitute an opportune time for interventions aimed not only at reducing anxiety symptoms but also at improving social skills for those with ASD, because such changes may result in increased independence, less avoidance and distress, and overall improvements in adaptive functioning during the adult years. The purpose of the present investigation was to evaluate the feasibility and efficacy of an adapted CBT protocol for addressing anxiety symptomatology and adaptive functioning in youth with comorbid ASD and anxiety disorder diagnoses.

With respect to clinicians' ratings of improvements, no participant was judged to have experienced a meaningful improvement in CSR (i.e., a decrease of more than 1 point) from the intake to postbaseline period by independent evaluators. After the intervention, all six participants were rated as exhibiting some improvement in CSR scores for both their primary and secondary diagnoses. CSR scores also remained fairly stable at follow-up (i.e., no CSR scores increased by more than 1 point), suggesting that clinician-rated improvements in overall symptom severity persisted following treatment.

Results were less clear when it came to interpreting parents' reports of their adolescent/young adult's anxiety symptoms. As described above, some participants were rated by their parent as evidencing spontaneous improvements during the baseline period, which made it difficult to draw conclusions about the effectiveness of the intervention. At posttreatment, five of the six participants were rated by their parents as evidencing some improvements on the BASC-2-PRS-A Anxiety and Internalizing Problems Composite scales, and these improvements resulted in statistically reliable RCI values for three of the six participants. It was notable that the one participant who did not evidence any change in symptoms was rated by his parent at intake and postbaseline assessments as having minimal levels of anxiety; hence, there was little room for improvement. Several participants were rated by parents as having some improvement on the RCADS-P Generalized Anxiety scale, although these changes were not statistically reliable as assessed by the RCI. Posttreatment, four of the six participants were judged to have experienced a decrease on the RCADS-P Social Anxiety scale, and this change represented a statistically reliable decrease for three of the participants. Taken together, these findings might suggest that parents viewed treatment as having a more beneficial effect on social anxiety symptoms compared to generalized worry.

With respect to parents' reports of ASD symptoms, results on the SRS Total scale were somewhat difficult to interpret as two of the six participants evidenced spontaneous improvements on the measure during the baseline period. At posttreatment, two participants evidenced no change on the SRS Total Score, one evidenced a statistically reliable worsening of symptoms, and two evidenced statistically reliable improvements on the SRS Total Score.

Spontaneous changes during the baseline period, both in the positive and negative directions, on the adolescents' BASC-2-SRP-A measure made it difficult to draw firm conclusions about the effectiveness of the intervention. Three adolescents (P1, P3, P5) reported an improvement in overall internalizing symptoms posttreatment, although this was statistically significant only for P3. At posttreatment, P2 reported a significant worsening of symptoms; however, this appeared to reflect a spontaneous remission of symptoms during postbaseline, and his final score posttreatment was actually lower than his score at intake. P4 evidenced no significant change posttreatment on either of the BASC-2-SRP-A Anxiety or Internalizing Problems Composite scales, although there was a slight worsening of Anxiety symptoms. Again, however, this appeared to be confounded by a large decrease in self-reported Anxiety symptoms postbaseline, and his final score posttreatment was lower than his intake score. The last participant, P6, evidenced significant reductions in both Anxiety and Internalizing Composite Problems scale scores postbaseline, and no significant changes at posttreatment.

Taken together these initial findings suggest that the AFIYA protocol was effective in decreasing clinician-rated levels of symptom severity and increasing clinician-rated levels of overall functioning. The intervention appeared somewhat effective in reducing parent-reported symptoms of anxiety, with reductions in social anxiety being more potent than reductions in generalized anxiety or worry. Unfortunately, the results did not appear to provide evidence that the protocol was effective across participants at reducing ASD symptoms, although these symptoms did improve for a subset of participants. Overall, results suggest that high-functioning adolescents and young adults with ASD may require additional practice with elements of certain modules to adequately address symptoms of worry and ASD—including the "C" module, ENTRY/EXIT module, and PLAY module—in order to generalize skills.

In regard to limitations of the current research, although the nature of the multiple-baseline design allows for substantive comparisons within and between subjects, our small sample size of six participants is certainly a limitation. Replication of this study with a larger sample size and randomized controlled group design would allow for more advanced statistical analyses of change over time. In addition, replication studies should recruit a more diverse sample, because the presence of only one female in the sample and lack of racial/ethnic diversity may limit generalizability of results of the current study. Current evidence from this trial with regard to clinician reports is compelling and suggests the possible efficacy of the approach. Therefore, such research with an RCT appears warranted. This is particularly true given the limited body of research on interventions for young adults with ASD and emotional disorder comorbidities to guide clinicians aiding these individuals.

It was also notable that the only participant (P6) who met criteria for a depressive disorder evidenced minimal improvement, relative to the other participants, during the intervention. Research indicates that anxiety and mood disorders often co-occur during adolescence (Garber & Weersing, 2010), and depression may be of increasing frequency among those with ASD as they age into adulthood (Ghaziuddin, Ghaziuddin, & Greden, 2002). To address this, the current protocol was adjusted to increase AFIYA's attention to depressive symptoms by adding modules on motivational enhancement and behavioral activation from an existing evidence-based treatment for emotional disorders in adolescence (Unified Protocols–Adolescence [UP-A]; Ehrenreich-May et al., 2018). However, additional attendance to the unique challenges of complex emotional disorder comorbidities may require additional modifications to the AFIYA protocol to produce change among older youth and young adults with behavioral withdrawal or mood involvement.

Despite these limitations, AFIYA appears to be a promising intervention for the treatment of anxiety in adolescents and young adults with ASD, a population for which there is currently a lack of efficacious treatments. Intervention during this time period is crucial, because the presence of an emotional disorder may exacerbate the unique challenges faced by individuals with ASD who are transitioning into new educational, occupational, and social settings. Further refinement of the protocol and replication of results with a larger sample size will help researchers and clinicians to better address the needs of this currently underserved population.

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